

Volume and Surface Area

1. The surface area of a cube is 384 cm^2 . Find its volume.

- A) 512 cm^3
- B) 216 cm^3
- C) 343 cm^3
- D) 729 cm^3

Answer: A) 512 cm^3

2. The ratio of the volumes of two cones is 1:4 and their diameters are in the ratio 4:5. The ratio of their heights is:

- A) 1:25
- B) 25:16
- C) 25:64
- D) 16:25

Answer: C) 25:64

3. A rectangular tank 15 m long and 10 m wide contains water up to a height of 4 m. What is the volume of water in kiloliters?

- A) 600 kL
- B) 400 kL
- C) 500 kL
- D) 800 kL

Answer: A) 600 kL

4. If the surface area of a sphere is 1386 cm^2 , then its volume is: (Use $\pi = 22/7$)

- A) 4851 cm^3
- B) 5841 cm^3
- C) 4581 cm^3
- D) 5481 cm^3

Answer: A) 4851 cm^3

5. The curved surface area of a cylinder is 880 cm^2 and its height is 20 cm. The volume of the cylinder is:

A) 2200 cm^3

B) 3080 cm^3

C) 4400 cm^3

D) 6160 cm^3

Answer: B) 3080 cm^3

6. The height of a cone is 15 cm and the diameter of its base is 16 cm. What is its total surface area? (Use $\pi = 3.14$)

A) 600.5 cm^2

B) 628 cm^2

C) 640 cm^2

D) 682 cm^2

Answer: B) 628 cm^2

7. A metallic sheet is of rectangular shape with dimensions $28 \text{ cm} \times 16 \text{ cm}$. From each of its four corners, a square is cut off so as to make an open box. If the length of the square is 3 cm, the volume of the box is:

A) 500 cm^3

B) 600 cm^3

C) 660 cm^3

D) 720 cm^3

Answer: C) 660 cm^3

8. The radii of two cylinders are in the ratio 3:5 and their heights are in the ratio 2:3. The ratio of their curved surface areas is:

A) 2:5

B) 3:5

C) 5:2

D) 5:3

Answer: A) 2:5

9. A hollow spherical shell is made of a metal of density 4.5 g/cm^3 . If its internal and external radii are 8 cm and 9 cm respectively, find the weight of the shell. (Use $\pi = 22/7$)

A) 3.84 kg

B) 4.22 kg

C) 4.88 kg

D) 5.12 kg

Answer: C) 4.88 kg

10. A cone of height 12 cm and radius 5 cm is carved out of a wooden block of the same dimensions. The percentage of wood wasted is:

A) 33.3%

B) 50%

C) 66.6%

D) 75%

Answer: C) 66.6%

11. A cuboidal water tank contains 216 liters of water. Its depth is $1/3$ of its length and breadth is $1/2$ of $1/3$ of the difference of length and depth. The length of the tank is:

A) 2 dm

B) 6 dm

C) 18 dm

D) 72 dm

Answer: C) 18 dm

12. The number of solid spheres, each of diameter 6 cm, that can be made by melting a solid metal cylinder of height 45 cm and diameter 4 cm is:

A) 3

B) 5

C) 6

D) 12

Answer: B) 5

13. The height of a cylinder is decreased by 50% and the radius is increased by 100%. The volume of the cylinder will:

- A) Increase by 100%
- B) Decrease by 50%
- C) Remain Unchanged
- D) Increase by 200%

Answer: A) Increase by 100%

14. A river 2.5 m deep and 45 m wide is flowing at the speed of 3.6 km/hour. The amount of water that runs into the sea per minute is:

- A) 6750 m³
- B) 6500 m³
- C) 6250 m³
- D) 6000 m³

Answer: A) 6750 m³

15. Find the length of the longest rod that can be placed in a room 16 m long, 12 m broad, and $\frac{32}{3}$ m high.

- A) 21 m
- B) 22 m
- C) 23 m
- D) $\frac{68}{3}$ m

Answer: D) $\frac{68}{3}$ m

16. The volume of a sphere is 4851 cm³. Its curved surface area is:

- A) 1286 cm²
- B) 1386 cm²
- C) 1486 cm²
- D) 1586 cm²

Answer: B) 1386 cm²

17. A solid cylinder has a total surface area of 231 cm^2 . Its curved surface area is $\frac{2}{3}$ of the total surface area. Find the volume of the cylinder.

A) 269.5 cm^3

B) 259.5 cm^3

C) 249.5 cm^3

D) 239.5 cm^3

Answer: A) 269.5 cm^3

18. If the areas of three adjacent faces of a cuboid are 12 m^2 , 20 m^2 , and 15 m^2 , then the volume of the cuboid is:

A) 300 m^3

B) 100 m^3

C) 80 m^3

D) 60 m^3

Answer: D) 60 m^3

19. A conical tent is to accommodate 11 persons. Each person must have 4 sq. meters of the space on the ground and 20 cubic meters of air to breathe. The height of the cone is:

A) 10 m

B) 12 m

C) 15 m

D) 18 m

Answer: C) 15 m

20. A spherical ball of radius 3 cm is melted and recast into three spherical balls. The radii of two of the balls are 1.5 cm and 2 cm. The radius of the third ball is:

A) 1.5 cm

B) 2 cm

C) 2.5 cm

D) 3 cm

Answer: C) 2.5 cm

21. The capacity of a cylindrical tank is 1848 m^3 and the diameter of its base is 14 m. The depth of the tank is:

- A) 8 m
- B) 10 m
- C) 12 m
- D) 14 m

Answer: C) 12 m

22. The surface area of a cube is 216 m^2 . Its diagonal is:

- A) 6 m
- B) $6\sqrt{3}$ m
- C) 12 m
- D) 18 m

Answer: B) $6\sqrt{3}$ m

23. Two cubes of edge 10 cm each are joined to form a single cuboid. The ratio of the total surface area of the new cuboid to the sum of the surface areas of the two cubes is:

- A) 1:1
- B) 2:3
- C) 5:6
- D) 4:5

Answer: C) 5:6

24. A well of diameter 3 m is dug 14 m deep. The earth taken out of it has been spread evenly all around it in the shape of a circular ring of width 4 m to form an embankment. Find the height of the embankment.

- A) 1.125 m
- B) 1.5 m
- C) 2.125 m
- D) 2.5 m

Answer: A) 1.125 m

25. A sphere of radius r is cut by a plane at a distance of h from its center, thereby breaking this sphere into two different pieces. The cumulative surface area of these two pieces is 25% more than that of the sphere. Find h .

A) $r / \sqrt{2}$

B) $r / 2$

C) $r / \sqrt{3}$

D) r

Answer: A) $r / \sqrt{2}$

26. A right circular cone is placed on a right circular cylinder of the same radius. The height of the cone is half that of the cylinder. If the total height of the solid is 9 cm and the radius of the base is 2 cm, find the volume of the solid.

A) $24\pi \text{ cm}^3$

B) $28\pi \text{ cm}^3$

C) $32\pi \text{ cm}^3$

D) $36\pi \text{ cm}^3$

Answer: B) $28\pi \text{ cm}^3$

27. If the radius of a sphere is doubled, its volume becomes:

A) 2 times

B) 4 times

C) 6 times

D) 8 times

Answer: D) 8 times

28. The number of coins of radius 0.75 cm and thickness 0.2 cm to be melted to make a right circular cylinder of height 8 cm and base radius 3 cm is:

A) 460

B) 540

C) 600

D) 640

Answer: D) 640

29. A cylinder, a cone, and a hemisphere are of equal base and have the same height. What is the ratio in their volumes?

A) 3 : 1 : 2

B) 3 : 2 : 1

C) 1 : 2 : 3

D) 1 : 3 : 2

Answer: A) 3 : 1 : 2

30. A cube of side 4 cm is cut into 1 cm cubes. What is the ratio of the surface area of the original cube to that of the sum of the surface areas of the smaller cubes?

A) 1:2

B) 1:4

C) 1:8

D) 1:16

Answer: B) 1:4

31. A rectangular sheet of metal 44 cm long and 20 cm broad is rolled along its length to form a cylinder. The volume of the cylinder so formed is:

A) 2880 cm^3

B) 3080 cm^3

C) 3280 cm^3

D) 3480 cm^3

Answer: B) 3080 cm^3

32. Three solid spheres of gold whose radii are 3 cm, 4 cm, and 5 cm respectively are melted into a single solid sphere. Find the radius of the sphere.

A) 5 cm

B) 6 cm

C) 7 cm

D) 8 cm

Answer: B) 6 cm

33. A solid wooden toy is in the shape of a right circular cone mounted on a hemisphere. If the radius of the hemisphere is 4.2 cm and the total height of the toy is 10.2 cm, find the volume of the wooden toy.

A) 266.11 cm^3

B) 300 cm^3

C) 366.11 cm^3

D) 400 cm^3

Answer: A) 266.11 cm^3

34. A hemispherical bowl is made of steel 0.25 cm thick. The inner radius of the bowl is 5 cm. Find the outer curved surface area of the bowl.

A) 163.25 cm^2

B) 173.25 cm^2

C) 183.25 cm^2

D) 193.25 cm^2

Answer: B) 173.25 cm^2

35. The volume of a wall, 5 times as high as it is broad and 8 times as long as it is high, is 12.8 m^3 . The breadth of the wall is:

A) 0.04 m

B) 0.4 m

C) 4.0 m

D) 40 m

Answer: B) 0.4 m

36. How many bricks, each measuring $25 \text{ cm} \times 12.5 \text{ cm} \times 7.5 \text{ cm}$, will be required to build a wall 6 m long, 5 m high and 0.5 m thick, while the mortar occupies 5% of the volume of the wall?

A) 5760

B) 6080

C) 6400

D) 7200

Answer: B) 6080

37. A sphere of radius 10.5 cm is melted and recast into small cones, each of radius 3.5 cm and height 3 cm. The number of such cones is:

- A) 106
- B) 116
- C) 126
- D) 136

Answer: C) 126

38. A right circular cylinder of radius r and height h ($h = 2r$) just encloses a sphere of diameter:

- A) r
- B) $2r$
- C) h
- D) $2h$

Answer: B) $2r$

39. The ratio of the total surface area to the lateral surface area of a cylinder with base radius 80 cm and height 20 cm is:

- A) 2:1
- B) 3:1
- C) 4:1
- D) 5:1

Answer: D) 5:1

40. If the areas of the three adjacent faces of a cuboid are x , y , z respectively, then the volume of the cuboid is:

- A) xyz
- B) $\sqrt{xyz}$
- C) $x^2y^2z^2$
- D) $(xyz)^2$

Answer: B) $\sqrt{xyz}$

41. A solid cylinder of brass 8 m high and 4 m diameter is melted and recast into a cone of diameter 3 m. The height of the cone is:

- A) 42.66 m
- B) 43.66 m
- C) 44.66 m
- D) 45.66 m

Answer: A) 42.66 m

42. A cone of height 9 cm with diameter of its base 18 cm is carved out from a wooden solid sphere of radius 9 cm. The percentage of wood wasted is:

- A) 25%
- B) 50%
- C) 75%
- D) 80%

Answer: C) 75%

43. A rectangular block of metal has dimensions 21 cm, 77 cm and 24 cm. The block is melted and a sphere is made. The radius of the sphere is:

- A) 14 cm
- B) 21 cm
- C) 28 cm
- D) 35 cm

Answer: B) 21 cm

44. The volume of a frustum of a cone of height 21 cm and radii of its circular ends 3 cm and 2 cm is:

- A) 134 cm^3
- B) 144 cm^3
- C) 154 cm^3
- D) 164 cm^3

Answer: C) 154 cm^3

45. Two solid cylinders of radii 4 cm and 5 cm and heights 6 cm and 4 cm respectively are melted and recast into a cylindrical disc of thickness 1 cm. The radius of the disc is:

- A) 7 cm
- B) 14 cm
- C) 21 cm
- D) 28 cm

Answer: B) 14 cm

46. A tent is in the form of a cylinder of diameter 20 m and height 2.5 m, surmounted by a cone of equal base and height 7.5 m. Find the capacity of the tent.

- A) 1450 m^3
- B) 1571.4 m^3
- C) 1600 m^3
- D) 1800 m^3

Answer: B) 1571.4 m^3

47. The slant height of a cone is increased by 10%. If the radius remains the same, the curved surface area increases by:

- A) 10%
- B) 12.5%
- C) 20%
- D) 21%

Answer: A) 10%

48. A cube of edge 6 cm is melted and smaller cubes of edge 2 cm are formed. How many such smaller cubes can be formed?

- A) 9
- B) 18
- C) 27
- D) 36

Answer: C) 27

49. Find the weight of a lead pipe 3.5 m long, if the external diameter of the pipe is 2.4 cm and the thickness of the lead is 2 mm and 1 cubic cm of lead weighs 11 gm.

- A) 3.6 kg
- B) 5.3 kg
- C) 6.8 kg
- D) 7.2 kg

Answer: B) 5.3 kg

50. The diameter of a roller is 84 cm and its length is 120 cm. It takes 500 complete revolutions to move once over to level a playground. Find the area of the playground in m^2 .

- A) 1584 m^2
- B) 1684 m^2
- C) 1784 m^2
- D) 1884 m^2

Answer: A) 1584 m^2

51. A metallic sphere of radius 10.5 cm is melted and then recast into small cones each of radius 3.5 cm and height 3 cm. The number of such cones is:

- A) 21
- B) 63
- C) 126
- D) 130

Answer: C) 126

52. The volume of a cube is 2744 cm^3 . Its surface area is:

- A) 196 cm^2
- B) 1176 cm^2
- C) 784 cm^2
- D) 588 cm^2

Answer: B) 1176 cm^2

53. A cistern of capacity 8000 liters measures externally 3.3 m by 2.6 m by 1.1 m and its walls are 5 cm thick. The thickness of the bottom is:

- A) 1 m
- B) 1 dm
- C) 1 cm
- D) 1.5 dm

Answer: B) 1 dm

54. The curved surface area of a right circular cone of height 15 cm and base diameter 16 cm is:

- A) $60\pi \text{ cm}^2$
- B) $68\pi \text{ cm}^2$
- C) $120\pi \text{ cm}^2$
- D) $136\pi \text{ cm}^2$

Answer: D) $136\pi \text{ cm}^2$

55. How many spherical bullets can be made out of a solid cube of lead whose edge measures 44 cm, each bullet being 4 cm in diameter?

- A) 2541
- B) 2550
- C) 2561
- D) 2575

Answer: A) 2541

56. The volume of a right circular cylinder is 1100 cm^3 and the radius of its base is 5 cm. Find its curved surface area.

- A) 220 cm^2
- B) 440 cm^2
- C) 550 cm^2
- D) 660 cm^2

Answer: B) 440 cm^2

57. A cylindrical vessel of diameter 9 cm has some water in it. A cylindrical iron piece of diameter 6 cm and height 6 cm is dropped into it. After it was completely immersed, the rise in the level of the water is:

- A) 1.6 cm
- B) 2.6 cm
- C) 3.6 cm
- D) 4.6 cm

Answer: B) 2.6 cm

58. A copper wire of 3 mm diameter is wound about a cylinder whose length is 1.2 m and diameter 10 cm, so as to cover the curved surface of the cylinder. Find the length of the wire.

- A) 125.7 m
- B) 130.4 m
- C) 135.2 m
- D) 140.6 m

Answer: A) 125.7 m

59. If the radius of a cylinder is halved and height is doubled, the volume will be:

- A) Same
- B) Doubled
- C) Halved
- D) Four times

Answer: C) Halved

60. The total surface area of a solid hemisphere of diameter 14 cm is:

- A) $147\pi \text{ cm}^2$
- B) $98\pi \text{ cm}^2$
- C) $196\pi \text{ cm}^2$
- D) $49\pi \text{ cm}^2$

Answer: A) $147\pi \text{ cm}^2$

61. A solid consists of a circular cylinder with an exact fitting sphere placed at the top. The height of the cylinder is equal to $2x$ and radius equal to x . The volume of the solid is:

- A) $3\pi x^3$
- B) $8\pi x^3/3$
- C) $7\pi x^3/3$
- D) $5\pi x^3/3$

Answer: B) $8\pi x^3/3$

62. A room is 15 feet long and 12 feet broad. The sum of the areas of the floor and the ceiling is equal to the sum of the areas of the four walls. The height of the room is:

- A) 5 ft
- B) 5.5 ft
- C) 6 ft
- D) 6.66 ft

Answer: D) 6.66 ft

63. The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be $1/27$ of the volume of the given cone, the height of the small cone is:

- A) 10 cm
- B) 15 cm
- C) 20 cm
- D) 25 cm

Answer: A) 10 cm

64. A parallelopiped whose sides are in ratio 2:4:8 have the same volume as a cube. The ratio of their surface area is:

- A) 7:5
- B) 4:3
- C) 7:6
- D) 8:5

Answer: C) 7:6

65. Water flows into a tank 150 m long and 100 m wide through a pipe of cross section 2 dm by 1.5 dm at a speed of 15 kmph. The time taken to reach 3 m depth is:

A) 100 hours

B) 110 hours

C) 120 hours

D) 130 hours

Answer: A) 100 hours

66. The number of cubes of side 3 cm that can be cut from a cube of side 6 cm is:

A) 4

B) 8

C) 27

D) 64

Answer: B) 8

67. A mason constructs a wall of dimensions 270 cm $\times$ 300 cm $\times$ 350 cm with bricks, each of size 22.5 cm $\times$ 11.25 cm $\times$ 8.75 cm. If it is assumed that $\frac{1}{8}$ of the space is covered by the mortar, then the number of bricks used to construct the wall is:

A) 11000

B) 11100

C) 11200

D) 11300

Answer: C) 11200

68. Find the volume of a sphere whose surface area is 154 cm².

A) 179.66 cm³

B) 200.33 cm³

C) 220.55 cm³

D) 250 cm³

Answer: A) 179.66 cm³

69. If the volumes of two cones are in the ratio 1:4 and their diameters are in the ratio 4:5, then the ratio of their heights is:

- A) 1:25
- B) 25:16
- C) 16:25
- D) 25:64

Answer: D) 25:64

70. A right circular cylinder and a sphere have the same radius and same volume. The ratio of their surface areas is:

- A) 3:2
- B) 4:3
- C) 7:6
- D) 14:13

Answer: C) 7:6

71. The capacity of a conical vessel with height 12 cm and slant height 13 cm is: (Use $\pi = 3.14$)

- A) 300 ml
- B) 314 ml
- C) 324 ml
- D) 334 ml

Answer: B) 314 ml

72. A cuboidal metal block of dimensions $20\text{ cm} \times 16\text{ cm} \times 12\text{ cm}$ weighs 6 kg. Find the weight of a block of the same metal of size $10\text{ cm} \times 8\text{ cm} \times 8\text{ cm}$.

- A) 700 g
- B) 800 g
- C) 900 g
- D) 1 kg

Answer: D) 1 kg

73. The base radius of a cylinder is 14 cm and its height is 30 cm. Find its total surface area.

A) 3872 cm^2

B) 3972 cm^2

C) 4072 cm^2

D) 4172 cm^2

Answer: A) 3872 cm^2

74. A hemisphere of lead of radius 6 cm is cast into a right circular cone of height 75 cm. The radius of the base of the cone is:

A) 1.4 cm

B) 2.4 cm

C) 2.8 cm

D) 4.2 cm

Answer: B) 2.4 cm

75. The ratio of the radii of two spheres is 1:2. The ratio of their surface areas is:

A) 1:2

B) 1:4

C) 1:8

D) 2:1

Answer: B) 1:4

76. A solid metallic cone of height 10 cm and radius of base 20 cm is melted to make spherical balls each of 4 cm diameter. How many such balls can be made?

A) 125

B) 250

C) 375

D) 500

Answer: A) 125

77. If the length of a rectangle is increased by 20% and breadth is decreased by 20%, then the area:

A) Increases by 4%

B) Decreases by 4%

- C) Remains same
- D) Increases by 10%

Answer: B) Decreases by 4%

78. The diagonals of three faces of a cuboid are x , y , z . Its volume is:

- A) xyz
- B) $\sqrt{x^2+y^2+z^2}$
- C) $\sqrt{(y^2+z^2-x^2)(z^2+x^2-y^2)(x^2+y^2-z^2)}/8$
- D) Undetermined

Answer: C) $\sqrt{(y^2+z^2-x^2)(z^2+x^2-y^2)(x^2+y^2-z^2)}/8$

79. A semicircular sheet of metal of diameter 28 cm is bent into an open conical cup. The depth of the cup is:

- A) 11 cm
- B) 12 cm
- C) 13 cm
- D) 12.12 cm

Answer: D) 12.12 cm

80. A cone measuring 21 cm in height and 8.4 cm in diameter is melted and recast into a sphere. The radius of the sphere is:

- A) 3.1 cm
- B) 4.2 cm
- C) 5.3 cm
- D) 6.4 cm

Answer: B) 4.2 cm

81. The height of a cylinder is 15 cm. The lateral surface area is 660 sq. cm. Its volume is:

- A) 1155 cm^3
- B) 2310 cm^3
- C) 3465 cm^3
- D) 4620 cm^3

Answer: B) 2310 cm^3

82. The volume of a cylinder is $448\pi \text{ cm}^3$ and height 7 cm. Find its lateral surface area.

A) $112\pi \text{ cm}^2$

B) $224\pi \text{ cm}^2$

C) $336\pi \text{ cm}^2$

D) $448\pi \text{ cm}^2$

Answer: A) $112\pi \text{ cm}^2$

83. Three cubes of iron whose edges are 6 cm, 8 cm and 10 cm respectively are melted and formed into a single cube. The edge of the new cube formed is:

A) 12 cm

B) 14 cm

C) 16 cm

D) 18 cm

Answer: A) 12 cm

84. The surface areas of two spheres are in the ratio 1:4. Then the ratio of their volumes is:

A) 1:4

B) 1:8

C) 1:16

D) 1:64

Answer: B) 1:8

85. A reservoir is in the shape of a frustum of a right circular cone. It is 8 m across at the top and 4 m across at the bottom. If it is 6 m deep, then its capacity is:

A) 176 m^3

B) 186 m^3

C) 196 m^3

D) 206 m^3

Answer: A) 176 m^3

86. A solid metal cylinder of radius 14 cm and height 21 cm is melted down and recast into spheres of radius 3.5 cm. The number of spheres that can be made is:

- A) 48
- B) 54
- C) 63
- D) 72

Answer: D) 72

87. A hall is 15 m long and 12 m broad. If the sum of the areas of the floor and the ceiling is equal to the sum of the areas of the four walls, the volume of the hall is:

- A) 720 m^3
- B) 900 m^3
- C) 1200 m^3
- D) 1800 m^3

Answer: C) 1200 m^3

88. If the surface area of a sphere is 346.5 cm^2 , then its radius is:

- A) 5.25 cm
- B) 5.5 cm
- C) 6.25 cm
- D) 6.5 cm

Answer: A) 5.25 cm

89. A right circular cone of radius 4 cm has a curved surface area of 47.1 cm^2 . Its slant height is:

- A) 3.5 cm
- B) 3.75 cm
- C) 4.0 cm
- D) 4.25 cm

Answer: B) 3.75 cm

90. A box is made of wood 3 cm thick. Its external length, breadth and height are 1.48 m, 1.16 m and 8.3 dm. Find the cost of painting the inner surface of the box at Rs. 50 per sq. m.

A) Rs. 270.50

B) Rs. 277.50

C) Rs. 280.50

D) Rs. 285.50

Answer: B) Rs. 277.50

91. Two cones have their heights in the ratio 1:3 and the radii of their bases in the ratio 3:1. The ratio of their volumes is:

A) 1:1

B) 1:3

C) 3:1

D) 9:1

Answer: C) 3:1

92. The slant height of a conical mountain is 2.5 km and the area of its base is 1.54 km^2 . The height of the mountain is:

A) 2.2 km

B) 2.3 km

C) 2.4 km

D) 2.5 km

Answer: C) 2.4 km

93. A solid metallic sphere of radius 8 cm is melted and recast into spherical balls each of radius 2 cm. The number of such balls is:

A) 32

B) 64

C) 128

D) 256

Answer: B) 64

94. The internal and external diameters of a hollow hemispherical vessel are 24 cm and 25 cm respectively. The total area to be painted is:

A) 1825.5 cm^2

B) 1925.5 cm^2

C) 2025.5 cm^2

D) 2125.5 cm^2

Answer: C) 2025.5 cm^2

95. The volume of a sphere is 38808 cm^3 . Its surface area is:

A) 5544 cm^2

B) 1386 cm^2

C) 4158 cm^2

D) 8316 cm^2

Answer: A) 5544 cm^2

96. If the radius of a cylinder is decreased by 20%, by what percent should the height be increased to maintain the same volume?

A) 20%

B) 36.25%

C) 56.25%

D) 60%

Answer: C) 56.25%

97. A cylindrical bucket 28 cm in diameter and 72 cm high is full of water. The water is emptied into a rectangular tank 66 cm long and 28 cm wide. Find the height of the water level in the tank.

A) 20 cm

B) 24 cm

C) 28 cm

D) 32 cm

Answer: B) 24 cm

98. The volume of the greatest sphere that can be cut off from a cylindrical log of wood of base radius 1 cm and height 5 cm is:

A) $\frac{4}{3} \pi \text{ cm}^3$

B) $\frac{10}{3} \pi \text{ cm}^3$

C) $5\pi \text{ cm}^3$

D) $\frac{20}{3} \pi \text{ cm}^3$

Answer: A) $\frac{4}{3} \pi \text{ cm}^3$

99. A cylinder and a cone have equal base radii and equal heights. If their curved surface areas are in the ratio 8:5, the ratio of radius to height is:

A) 3:4

B) 3:5

C) 4:3

D) 5:3

Answer: A) 3:4

100. A tank 40 m long, 30 m broad and 12 m deep is dug in a field 1000 m long and 30 m broad. By how much will the level of the field rise if the earth dug out of the tank is evenly spread over the field?

A) 0.5 m

B) 0.533 m

C) 1.5 m

D) 2 m

Answer: B) 0.533 m